Lecture 3: Growth Accounting, Development Accounting, and Misallocation

Macroeconomics I — Exam Reference Cheatsheet

BSE Macroeconomics I

Contents

1 Notation and Definitions 2

2 Growth Accounting 3

3 Development Accounting — Hall & Jones (1999) 6

4 Misallocation — Hsieh & Klenow (2009) 9

5 Problem-Solving Templates 13

6 Common Pitfalls 14

7 Quick Reference Formulas 15

1 Notation and Definitions

Core Variables

Y_t = aggregate output

K_t = capital stock

L_t = labor (workers)

N_t = population

A_t = Total Factor Productivity (TFP)

B_t = labor-augmenting technology

$H_i = e^{\phi(E_i)} L_i$ = human-capital-adjusted labor

$h_i \equiv H_i / L_i$

α = capital share ($\approx 1/3$)

δ = depreciation rate

σ = CES elasticity of substitution

τ_i^Y, τ_i^K = output/capital wedges

Growth Rate Notation

$$\gamma_X \equiv \frac{\dot{X}}{X} = \frac{d \ln X}{dt} \qquad \text{(continuous time)}$$

Discrete time: $\gamma_X \approx \ln X_{t+1} - \ln X_t$.

Key Identities

$$y_i \equiv Y_i / L_i \quad \text{(output per worker)}$$
$$k_i \equiv K_i / L_i \quad \text{(capital per worker)}$$
$$K/Y = \text{capital-output ratio (a key observable)}$$

2 Growth Accounting

The Core Idea

Growth accounting uses a production function to decompose observed output growth into contributions from capital, labor, and a residual (TFP). It is purely *accounting* — it does not identify causal relationships.

Primal Approach: Log-Differentiation Derivation

Step 1. Start from a general production function:

$$Y_t = A_t F(K_t, L_t)$$

Step 2. Take natural logarithms:

$$\ln Y_t = \ln A_t + \ln F(K_t, L_t)$$

Step 3. Differentiate with respect to time (log-differentiation):

$$\frac{\dot{Y}}{Y} = \frac{\dot{A}}{A} + \frac{F_K K}{F}\frac{\dot{K}}{K} + \frac{F_L L}{F}\frac{\dot{L}}{L}$$

Step 4. Identify factor shares. Under perfect competition with CRS, a profit-maximizing firm sets $F_K p = r$ and $F_L p = w$, so:

$$\frac{F_K K}{F} = \frac{rK}{pY} \equiv \alpha_t \qquad \text{(capital income share)}$$

$$\frac{F_L L}{F} = \frac{wL}{pY} \equiv 1 - \alpha_t \qquad \text{(labor income share)}$$

Step 5. Substitute:

$$\gamma_Y = \gamma_A + \alpha_t \gamma_K + (1 - \alpha_t)\gamma_L$$

Step 6. Since γ_A is the only unobservable, solve as a residual:

$$\gamma_A = \gamma_Y - \alpha_t \gamma_K - (1 - \alpha_t)\gamma_L$$

$$\boxed{\gamma_Y = \gamma_A + \alpha_t \gamma_K + (1 - \alpha_t)\gamma_L} \qquad \text{(Growth Decomposition)}$$

$$\boxed{\gamma_A = \gamma_Y - \alpha_t \gamma_K - (1 - \alpha_t)\gamma_L} \qquad \text{(Solow Residual)}$$

With Human Capital (B_t)

The alternative framework uses $Y_t = K_t^{\alpha}(B_t L_t)^{1-\alpha}$, which rearranges to:

$$\frac{Y_t}{N_t} = B_t \frac{L_t}{N_t}\left(\frac{K_t}{Y_t}\right)^{\frac{\alpha}{1-\alpha}}$$

Taking growth rates:

$$\gamma_{Y/N} = \gamma_B + \gamma_{L/N} + \frac{\alpha}{1-\alpha}\gamma_{K/Y} \tag{Alt. Decomposition}$$

This decomposes growth of output per capita into:

- γ_B: growth of the Solow residual (labor-augmenting TFP)
- $\gamma_{L/N}$: growth of labor force participation per capita
- $\frac{\alpha}{1-\alpha}\gamma_{K/Y}$: capital deepening (change in capital-output ratio)

US Data (Jones, 2015): 1948–2013 average growth $Y/N = 2.5\%$ per year, of which $B \approx 2.0\%$, $L/N \approx 0.3\%$, $K/Y \approx 0.1\%$. Capital deepening is small — the US is near its BGP.

Dual Approach

Starting from the factor payments identity $Y_t = w_t L_t + r_t K_t$, log-differentiating gives:

$$\gamma_A = \alpha_K \frac{\dot{r}}{r} + \alpha_L \frac{\dot{w}}{w} \tag{Dual TFP}$$

Primal vs. Dual: Should coincide in theory. In practice they differ (Hsieh, 2002). Notably, Singapore's primal TFP growth was -0.7% but dual was $+2.2\%$, raising concerns about capital quality mismeasurement.

What Is Inside γ_A?

The Solow residual is a *residual* — it captures:

- Genuine technological progress
- Misallocation (market inefficiency, cronyism, regulation)
- Mismeasurement of factor quality (especially capital and education)

Historical finding: Early studies attributed ∼70% of US growth to TFP. Better capital quality measurement reduces this to 1/3–1/2.

Factor Shares vs. Elasticities: When Does It Work?

The substitution $\alpha_t = rK/pY$ for $F_K K/Y$ requires:

1. **Perfect competition:** firms equate revenue MPK to input price. With market power (ϵ_D finite), cost share $\neq$ income share.
2. **No fixed costs:** only variable costs appear in factor shares.
3. **No externalities:** private and social marginal products coincide.

Measuring Capital: Perpetual Inventory Method

$$\boxed{K_{t+1} = (1-\delta)K_t + I_t} \tag{PIM}$$

Initial condition (Hall-Jones): $K_0 = \dfrac{I_0}{g+\delta}$ (assumes economy on BGP at start).

Quality adjustment is critical: a computer today delivers far more computational capital than 30 years ago at the same nominal price. Use *quality-adjusted* investment deflators.

3 Development Accounting — Hall & Jones (1999)

Goal

Decompose cross-country income differences into contributions from:

1. Physical capital accumulation
2. Human capital accumulation
3. TFP / productivity (A_i)

Production Function Setup

Hall and Jones use:

$$Y_i = K_i^\alpha (A_i H_i)^{1-\alpha}, \qquad H_i = e^{\phi(E_i)} L_i$$

where E_i = years of schooling and $\phi(\cdot)$ is the return-to-schooling function (piecewise linear, estimated from Mincerian wage regressions).

Step 1. Divide both sides by L_i:

$$y_i = \frac{Y_i}{L_i} = \frac{K_i^\alpha (A_i H_i)^{1-\alpha}}{L_i}$$

Step 2. Rewrite $K_i^\alpha = (K_i/Y_i)^\alpha Y_i^\alpha$. Then:

$$y_i = K_i^\alpha (A_i h_i)^{1-\alpha} \qquad h_i \equiv H_i/L_i$$

Step 3. Use CRS to factor out y_i^α:

$$y_i = y_i^\alpha \left(\frac{K_i}{Y_i}\right)^\alpha (A_i h_i)^{1-\alpha}$$

Step 4. Solve for $y_i^{1-\alpha}$, then raise to $1/(1-\alpha)$:

$$y_i = \left(\frac{K_i}{Y_i}\right)^{\frac{\alpha}{1-\alpha}} h_i A_i$$

$$\boxed{y_i = \left(\frac{K_i}{Y_i}\right)^{\frac{\alpha}{1-\alpha}} h_i A_i} \qquad \text{(HJ Decomposition)}$$

Taking logs and differencing across countries (relative to US):

$$\ln \frac{y_i}{y_{US}} = \frac{\alpha}{1-\alpha} \ln \frac{(K/Y)_i}{(K/Y)_{US}} + \ln \frac{h_i}{h_{US}} + \ln \frac{A_i}{A_{US}}$$

Measuring A_i as a Residual

Since y_i, K_i/Y_i, and h_i are all measurable, TFP is recovered as:

$$A_i = \frac{y_i}{\left(\frac{K_i}{Y_i}\right)^{\frac{\alpha}{1-\alpha}} h_i} \qquad \text{(TFP Residual)}$$

with $\alpha = 1/3$ (standard capital share).

Human Capital Function

Mincerian piecewise returns to schooling used by Hall & Jones:

$$\phi(s) = \begin{cases} 0.134 \times s & s \leq 4 \\ 0.134 \times 4 + 0.101 \times (s-4) & 4 < s \leq 8 \\ 0.134 \times 4 + 0.101 \times 4 + 0.068 \times (s-8) & s > 8 \end{cases}$$

Key Results

Country	Y/L	$(K/Y)^{\alpha/(1-\alpha)}$	H/L	A
United States	1.000	1.000	1.000	1.000
West Germany	0.818	1.118	0.802	0.912
Japan	0.587	1.119	0.797	0.658
Mexico	0.433	0.868	0.538	0.926
India	0.086	0.709	0.454	0.267
China	0.060	0.891	0.632	0.106
Kenya	0.056	0.747	0.457	0.165
Average (127)	0.296	0.853	0.565	0.516

Headline decomposition (top 5 vs. bottom 5 countries):

$$y_{\text{top 5}} = 1.8 \underbrace{}_{\text{Capital}} \times 2.2 \underbrace{}_{\text{Human Cap.}} \times 8.3 \underbrace{}_{\text{TFP}} \times y_{\text{bottom 5}}$$

Main finding: TFP differences account for **50–60%** of cross-country income variance (Caselli, 2005). Factor accumulation alone cannot explain why rich countries are so much richer than poor ones.

"Success" Criterion for Development Accounting

The variance-based success criterion: what fraction of $\text{Var}(\ln y_i)$ is explained by factor accumulation alone?

$$\text{"Success"} = \frac{\text{Var}(\ln \hat{y}_i)}{\text{Var}(\ln y_i)}, \qquad \hat{y}_i \equiv \left(\frac{K_i}{Y_i}\right)^{\frac{\alpha}{1-\alpha}} h_i$$

Caselli (2005) finds: whole sample $\approx$ 38% explained by factors $\Rightarrow$ TFP explains $\sim$ 62%. Non-OECD: factors explain only $\sim$ 36%.

Social Infrastructure and Institutions (Hall-Jones)

Hall and Jones relate TFP to "social infrastructure" S_i (GADP index: law and order, anti-corruption, contract enforcement):

$$\log y_i = \gamma + \beta S_i + \varepsilon_i$$

Problem: **reverse causality** (higher income enables better institutions). Solution: IV with instruments uncorrelated with current y_i:

- Distance from equator
- Fraction of population speaking Western language
- Frankel-Romer predicted trade share

Result: $\hat{\beta} \approx 5.1$ (main IV specification), significant — social infrastructure strongly predicts income levels.

4 Misallocation — Hsieh & Klenow (2009)

Motivation: What Is Inside A?

TFP is a black box. One major component is **misallocation**: an inefficient allocation of resources across firms that lowers aggregate output even holding total inputs fixed.

When Does Allocation NOT Matter? (Null Result)

Suppose all firms: (i) have the same productivity $A_i = A$; (ii) face common factor prices r, w; (iii) are perfectly competitive in a homogeneous goods market.

Then all firms choose the same k/l ratio:

$$\frac{k_i^*}{l_i^*} = \frac{\alpha}{1-\alpha}\frac{w}{r} \quad \forall i$$

Aggregating: $Y = \int_i k_i^\alpha l_i^{1-\alpha} = K^\alpha L^{1-\alpha}$. The distribution of resources across identical firms is irrelevant.

Hsieh-Klenow Model Setup

Households: CES preferences over differentiated varieties:

$$C = \left(\int_i c_i^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1$$

Demand: $c_i = p_i^{-\sigma} P^\sigma C$.

Firms: Monopolistically competitive, heterogeneous productivity A_i:

$$y_i = A_i k_i^\alpha l_i^{1-\alpha} \qquad \text{(Firm Technology)}$$

Undistorted equilibrium pricing:

$$p_i = \frac{\sigma}{\sigma-1}\frac{x}{A_i}, \qquad x \equiv \left(\frac{r}{\alpha}\right)^\alpha \left(\frac{w}{1-\alpha}\right)^{1-\alpha}$$

In the efficient benchmark all $\text{MRPK}_i = r$ and $\text{MRPL}_i = w$: marginal revenue products are equalized and $y_i \propto A_i^\sigma$.

Introducing Distortions (Wedges)

Firm i now faces an output wedge τ_i^Y and a capital wedge τ_i^K:

$$\max_{y_i, k_i, l_i} (1-\tau_i^Y) p_i(y_i) y_i - w l_i - (1+\tau_i^K) r k_i$$

FOC w.r.t. l_i:

$$(1-\tau_i^Y)\frac{\partial(p_i y_i)}{\partial l_i} = w \implies \text{MRPL}_i = \frac{w}{1-\tau_i^Y}$$

FOC w.r.t. k_i:

$$(1-\tau_i^Y)\frac{\partial(p_i y_i)}{\partial k_i} = (1+\tau_i^K)r \implies \text{MRPK}_i = \frac{r(1+\tau_i^K)}{1-\tau_i^Y}$$

Distorted k/l ratio:

$$\frac{k_i}{l_i} = \frac{\alpha}{1-\alpha}\frac{w}{r}\frac{1}{1+\tau_i^K}$$

Distorted output share:

$$y_i \propto A_i^\sigma \left(\frac{1-\tau_i^Y}{1+\tau_i^K}\right)^\sigma$$

Under distortions, MRPs **differ across firms**:

$$\boxed{\text{MRPK}_i = \frac{r(1+\tau_i^K)}{1-\tau_i^Y}} \qquad \boxed{\text{MRPL}_i = \frac{w}{1-\tau_i^Y}} \qquad \text{(Distorted MRPs)}$$

Dispersion in MRPK_i (or equivalently in TFPR_i) is the empirical signature of misallocation.

TFPR vs. TFPQ

Define two TFP concepts:

$$\text{TFPQ}_i \equiv A_i \quad \text{(physical TFP — "how productive is the firm?")}$$
$$\text{TFPR}_i \equiv p_i A_i \quad \text{(revenue TFP — "how much revenue per unit of input?")}$$

In the efficient equilibrium: $\text{TFPR}_i = PA$ is constant across all firms.
With distortions:

$$\text{TFPR}_i = \frac{\sigma}{\sigma-1} x \frac{(1+\tau_i^K)^\alpha}{1-\tau_i^Y}$$

Higher TFPR_i indicates the firm faces binding constraints (high τ_i^K or low $1-\tau_i^Y$) relative to its productivity.

Aggregate TFP with Distortions

Aggregate output: $Y = AK^{\alpha}L^{1-\alpha}$ where aggregate TFP is:

$$A = \left[\int_i \left(A_i \frac{\overline{\text{TFPR}}}{\text{TFPR}_i}\right)^{\sigma-1}\right]^{\frac{1}{\sigma-1}}$$

where $\overline{\text{TFPR}} = PA$ is the aggregate TFPR.
Since the aggregator is **concave** in A_i/TFPR_i, Jensen's inequality implies:

$$A \leq A_{\text{efficient}} = \left[\int_i A_i^{\sigma-1}\right]^{\frac{1}{\sigma-1}}$$

with equality only when TFPR_i is constant across all firms.
Greater dispersion in $\text{TFPR}_i \Rightarrow$ lower aggregate A.

$$\boxed{A = \left[\int_i \left(A_i \frac{\overline{\text{TFPR}}}{\text{TFPR}_i}\right)^{\sigma-1}\right]^{\frac{1}{\sigma-1}} \leq A_{\text{efficient}}} \qquad \text{(Aggregate TFP)}$$

Intuition: Misallocation acts like a tax on high-productivity firms. Resources flow to low-productivity, low-wedge firms. The aggregate TFP loss equals the welfare cost of this distorted allocation.

Factor Shares Under Monopolistic Competition

Because firms charge a markup $\sigma/(\sigma-1)$ over marginal cost, measured income shares no longer equal output elasticities:

$$\Theta_K = \alpha\frac{\sigma-1}{\sigma} < \alpha, \qquad \Theta_L = (1-\alpha)\frac{\sigma-1}{\sigma} < 1-\alpha$$

Counterfactual: TFP Gains from Eliminating Misallocation

HK ask: what would TFP be if all TFPR_i were equalized within industries?

Country	Early year	Mid year	Late year
China	115.1% (1998)	95.8% (2001)	86.6% (2005)
India	100.4% (1987)	102.1% (1991)	127.5% (1994)
United States	36.1% (1977)	30.7% (1987)	42.9% (1997)

Policy implication: Equalizing TFPR within industries would raise Chinese manufacturing TFP by ∼87–115%, Indian TFP by ∼100–128%, relative to the US gain of ∼31–43%. Even relative to US gains, India and China have substantially more misallocation.

Caveats to HK Interpretation

Measured TFPR dispersion could also reflect:

- Measurement error in prices and quantities
- Unmodeled adjustment costs (lumpy investment, entry/exit)
- Deviations from CES/CD/CRS assumptions
- Dynamic considerations (learning-by-doing, R&D investment)

These can overturn the conclusion that all TFPR dispersion is welfare-reducing.

5 Problem-Solving Templates

Template A: Growth Accounting Calculation

Given: time-series data on Y, K, L, and α.

1. Compute growth rates: $\gamma_Y = \Delta \ln Y$, $\gamma_K = \Delta \ln K$, $\gamma_L = \Delta \ln L$
2. Apply Solow residual formula:
$$\gamma_A = \gamma_Y - \alpha\gamma_K - (1-\alpha)\gamma_L$$
3. Decompose as shares of γ_Y:
$$1 = \underbrace{\frac{\alpha\gamma_K}{\gamma_Y}}_{\text{capital contrib.}} + \underbrace{\frac{(1-\alpha)\gamma_L}{\gamma_Y}}_{\text{labor contrib.}} + \underbrace{\frac{\gamma_A}{\gamma_Y}}_{\text{TFP contrib.}}$$
4. Interpret: if γ_A/γ_Y is large ($> 1/2$), growth is TFP-driven.

Template B: Development Accounting (Hall-Jones Style)

Given: cross-country data on y_i, K_i/Y_i, E_i (years of schooling).

1. Compute human capital: $h_i = e^{\phi(E_i)}$ (piecewise Mincer function)
2. Compute factor component: $\hat{y}_i = (K_i/Y_i)^{\alpha/(1-\alpha)} h_i$
3. Recover TFP as residual: $A_i = y_i/\hat{y}_i$
4. Compute success criterion: $\mathrm{Var}(\ln \hat{y}_i)/\mathrm{Var}(\ln y_i)$
5. Interpret: low success ratio $\Rightarrow$ TFP differences dominate

Template C: Misallocation Analysis (HK Style)

Given: firm-level data on revenue $p_i y_i$, capital k_i, labor l_i.

1. Compute revenue productivity: $\mathrm{TFPR}_i = p_i A_i \propto (p_i y_i)/(k_i^\alpha l_i^{1-\alpha})$
2. Compute MRPs:
$$\mathrm{MRPK}_i = \alpha \frac{p_i y_i}{k_i}, \qquad \mathrm{MRPL}_i = (1-\alpha)\frac{p_i y_i}{l_i}$$
3. Measure dispersion: $\mathrm{Var}(\ln \mathrm{MRPK}_i)$ or $\mathrm{Var}(\ln \mathrm{TFPR}_i)$ within industries
4. Counterfactual: set $\mathrm{TFPR}_i = \overline{\mathrm{TFPR}}$ for all i, compute new A
5. TFP gain $= A_{\text{efficient}}/A_{\text{actual}} - 1$

Template D: Distinguishing Growth Accounting from Development Accounting

	Growth Accounting	**Development Accounting**
Data	Time series for one country	Cross-section of countries
Question	Why did country i grow?	Why is country i richer than j?
Equation	$\Delta \ln Y = \gamma_A + \alpha\gamma_K + (1-\alpha)\gamma_L$	$\ln(y_R/y_P) = \ln(A_R/A_P) + \frac{\alpha}{1-\alpha}\ln(K/Y)_R/(K/Y)_P + \ldots$
Key finding	TFP accounts for 1/3–1/2 of growth	TFP accounts for ∼50–60% of income differences

6 Common Pitfalls

Growth accounting is a **time-series** exercise for a single country. Development accounting is a **cross-sectional** exercise across countries. The formulas look similar, but the data, the residuals, and the interpretation differ.

γ_A is a **residual**, not a direct measurement. It captures everything we do not measure: technology, misallocation, measurement error. Saying "TFP growth was 2%" does NOT mean "technological progress was 2%".

Even if capital accounts for 50% of growth, this does not mean capital was the *cause*. On a BGP, all capital accumulation is itself driven by technology: without $\gamma_A > 0$, capital per effective worker would not grow.

The identity $\alpha_t = rK/pY = F_K K/Y$ requires perfect competition. With monopoly power or fixed costs, measured factor shares understate the true elasticities. See: cost shares vs. income shares.

$\text{TFPQ}_i = A_i$ is physical productivity (units of output per unit of inputs). $\text{TFPR}_i = p_i A_i$ is revenue productivity. In the efficient equilibrium, TFPR is equalized across firms, not TFPQ. High-productivity firms have lower prices but the same TFPR. Misallocation shows up in TFPR dispersion, not TFPQ dispersion.

When TFP is measured as a residual ($A_i = y_i/\hat{y}_i$), we have not *explained* TFP differences — we have merely *labeled* the unexplained portion. Explaining A_i requires a theory (e.g., institutions, as in Hall-Jones).

The capital term in the per-worker decomposition is $(K/Y)^{\alpha/(1-\alpha)}$, **not** $(K/Y)^{\alpha}$. This arises from algebraic manipulation of the Cobb-Douglas function. With $\alpha = 1/3$, the exponent is $1/2$.

7 Quick Reference Formulas

Growth Accounting

$$\gamma_Y = \gamma_A + \alpha\gamma_K + (1-\alpha)\gamma_L \qquad \text{(Primal)}$$
$$\gamma_A = \gamma_Y - \alpha\gamma_K - (1-\alpha)\gamma_L \qquad \text{(Solow Residual)}$$
$$\gamma_A = \alpha_K \dot{r}/r + \alpha_L \dot{w}/w \qquad \text{(Dual)}$$
$$\gamma_{Y/N} = \gamma_B + \gamma_{L/N} + \tfrac{\alpha}{1-\alpha}\gamma_{K/Y} \qquad \text{(Alt. with } K/Y \text{ ratio)}$$

Capital Measurement

$$K_{t+1} = (1-\delta)K_t + I_t \qquad \text{(Perpetual Inventory)}$$
$$K_0 = I_0/(g+\delta) \qquad \text{(Initial condition, on BGP)}$$

Development Accounting (Hall-Jones)

$$y_i = \left(\frac{K_i}{Y_i}\right)^{\frac{\alpha}{1-\alpha}} h_i A_i \qquad \text{(Output per worker)}$$
$$h_i = e^{\phi(E_i)} \qquad \text{(Human capital, Mincer)}$$
$$A_i = y_i \Big/ \left[\left(\frac{K_i}{Y_i}\right)^{\frac{\alpha}{1-\alpha}} h_i\right] \qquad \text{(TFP residual)}$$
$$\ln\frac{y_R}{y_P} = \underbrace{\tfrac{\alpha}{1-\alpha}\ln\frac{(K/Y)_R}{(K/Y)_P}}_{\text{capital}} + \underbrace{\ln\frac{h_R}{h_P}}_{\text{human cap.}} + \underbrace{\ln\frac{A_R}{A_P}}_{\text{TFP}}$$

Misallocation (Hsieh-Klenow)

$$\text{MRPK}_i = \frac{r(1+\tau_i^K)}{1-\tau_i^Y} \qquad \text{MRPL}_i = \frac{w}{1-\tau_i^Y}$$
$$\text{TFPR}_i = p_i A_i = \frac{\sigma}{\sigma-1} x \frac{(1+\tau_i^K)^\alpha}{1-\tau_i^Y} \qquad y_i \propto A_i^\sigma \left(\frac{1-\tau_i^Y}{1+\tau_i^K}\right)^\sigma$$
$$A = \left[\int_i \left(\frac{A_i \overline{\text{TFPR}}}{\text{TFPR}_i}\right)^{\sigma-1}\right]^{\frac{1}{\sigma-1}} \qquad \leq A_{\text{efficient}}$$

Key Parameter Values (Standard Calibration)

Parameter	Value	Source
α (capital share)	1/3	National accounts
δ (depreciation)	0.06	Hall & Jones (1999)
$\phi'(s)$ for first 4 years	0.134	Mincerian regressions
$\phi'(s)$ for years 5–8	0.101	Mincerian regressions
$\phi'(s)$ for year 9+	0.068	Mincerian regressions

Key Empirical Findings Summary

Finding	**Implication**
TFP explains ∼50–60% of cross-country income variation	Factor accumulation alone cannot explain global inequality
Primal $\neq$ Dual TFP (Singapore)	Capital quality mismeasurement is important
China/India TFP gain from equalization ∼100%	Severe misallocation in developing countries
Top/bottom 5 countries: A ratio = 8.3	Most income gap is in productivity, not factors